

INDEPENDENT PROTOTYPE · SYNTHETIC DATA

VC Retail Analytics

An independent analytics prototype for designer-lighting retail workflows. Synthetic retail data sized to public Visual Comfort scale, controlled metric queries, voice on Snowflake, Market / competitor share plays, and an Interview Lab for AI-engineering conversation.

The typed assistant is **not an LLM**. It routes supported prompts to approved metric queries and returns traceable SQL and rows. Voice can fuzzy-route, then fall back to Cortex Analyst on Snowflake. That constraint is the engineering point: tool access, metric definitions, user scope, and testable responses.

Open hosted demo →

<https://vc.datasharkbi.com> [GitHub \(public\)](#)

SYNTHETIC

SQLITE + SNOWFLAKE

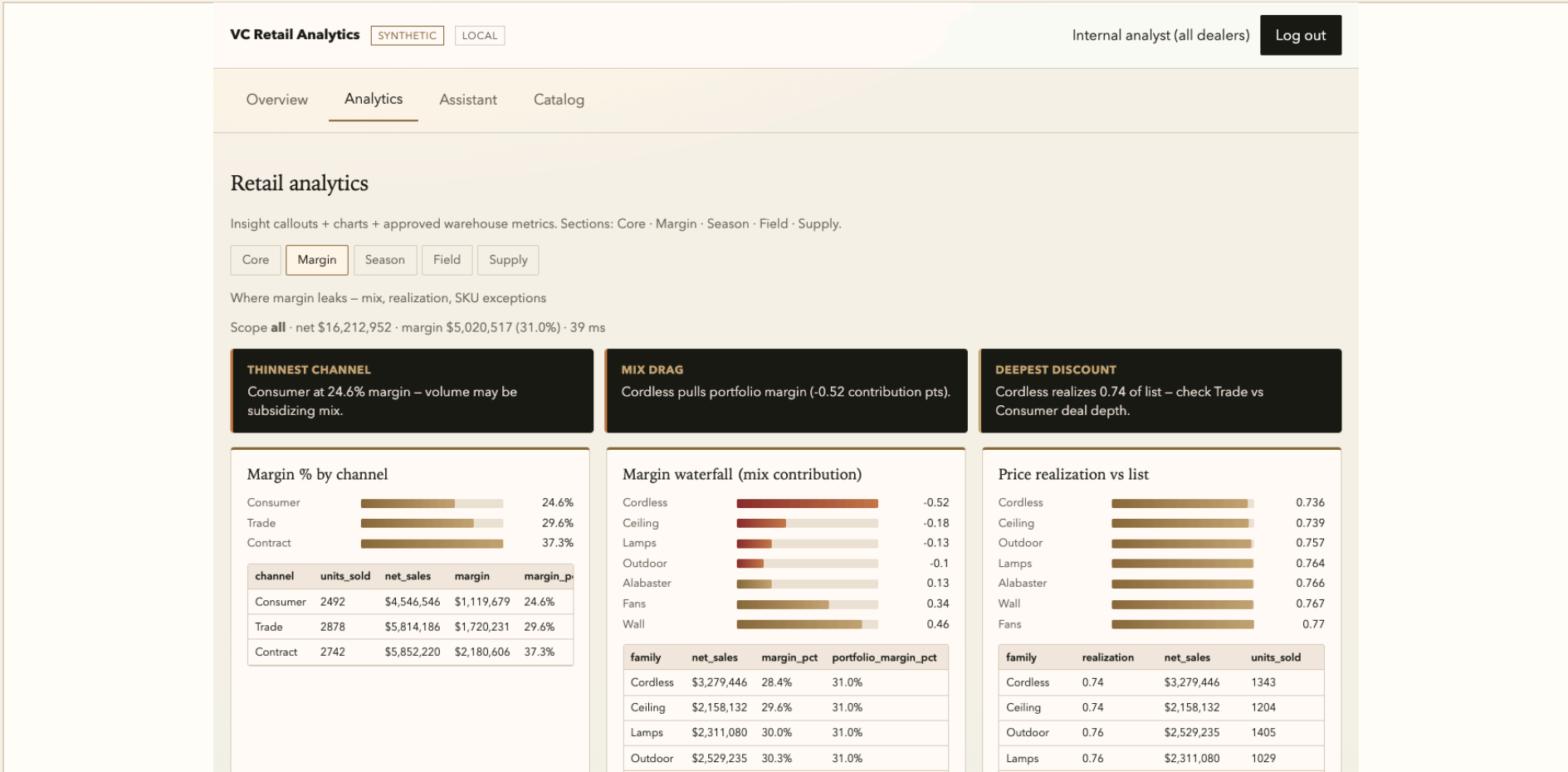
INTERVIEW LAB

ACCESS-GATED

Paul Rendleman · independent prototype

Which channel has the thinnest margin?

Live capture from the synthetic app (operator session). Margin section shows margin % by channel, mix contribution waterfall, and price realization. Insight callouts summarize the approved metric results.



Supported prompt → approved query → result

VC Retail Analytics

SYNTHETICLOCAL

Internal analyst (all dealers)Log out

OverviewAnalyticsAssistantCatalog

Illustrative support vs analytics workflow

Left: illustrative FAQ / escalate pattern. Right: deterministic assistant that routes exact prompts to approved metric queries (not an LLM). Illustrative workflows, not an assessment of any client's internal systems.

Illustrative support workflow

e.g. custom order lead time

Ask support pattern

Analytics workflow (demonstrated)

show margin percent

Ask approved metricEscalate packet

offline governed assistant (not LLM) - margin_pct

Thinnest channel in result set: Consumer at 24.6% margin.

Margin percent by channel (margin \$ / net sales).

Consumer24.6%

Trade29.6%

Contract37.3%

channel	units_sold	net_sales	margin	margin_pct
Consumer	2492	\$4,546,546	\$1,119,679	24.6%
Trade	2878	\$5,814,186	\$1,720,231	29.6%
Contract	2742	\$5,852,220	\$2,180,606	37.3%

► SQL + tool trace

► Supported metric prompts (exact match)

Exact prompt used

show margin percent

What returned

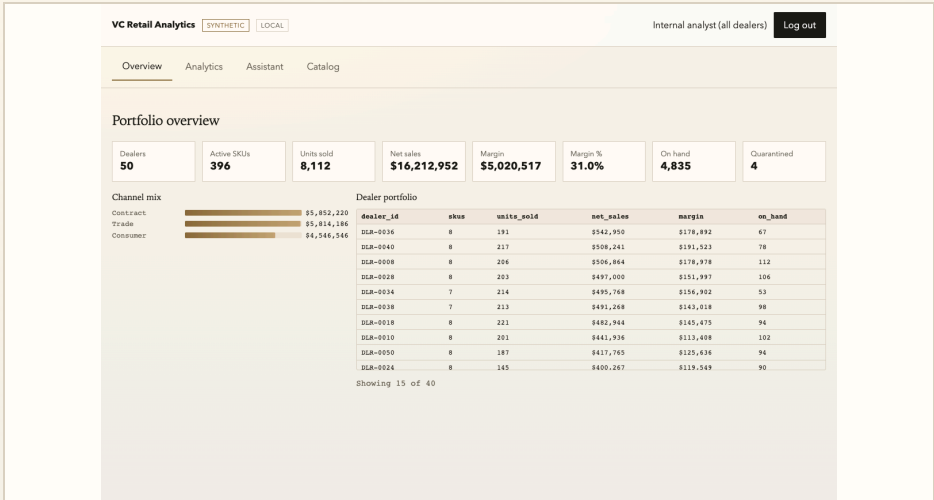
Metric `margin_pct`, one-line insight, bar chart, table, and expandable SQL + tool trace. Scope follows the signed session (operator = all dealers).

Unsupported requests

Invented forecasts and free-form questions are refused or clarified. The router only accepts an exact supported vocabulary.

Proposed (not built): an LLM could interpret natural language, then call the same approved tools, with authorization enforced outside the model.

Analytics sections actually implemented



Illustrative rule-based scores

Rep A-D grades and vendor Prefer / Watch / Exit are synthetic, rule-based composites (attainment, margin vs peer, mix, OTIF, quality). They are demo decision aids, not real personnel or supplier recommendations.

Market

Demand outlook, competitor landscape (public-estimate), share expansion, GM lift, growth momentum.

Margin

Margin %, realization, mix waterfall, low-margin SKUs.

Season & Field

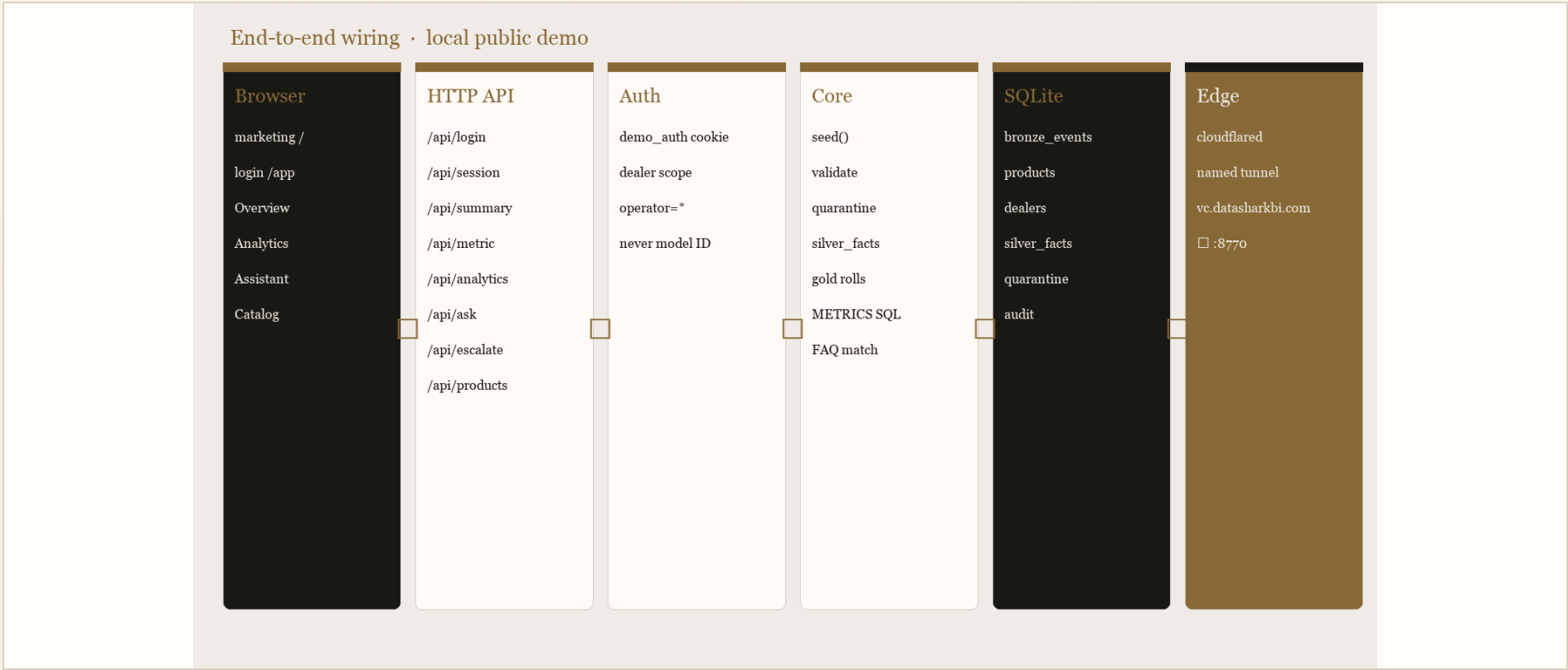
Seasonal index, YoY, territories, whitespace, rep grades.

Supply & Plan

Days of cover, reorder, vendor Prefer/Watch/Exit, MRP, forecast accuracy.

Analytics subnav includes Market · Core · Margin · Season · Field · Supply · Procurement · Planning. Interview Lab adds board brief, golden evals, Cortex vs governed, RAP-style scope switch, and a 90-day plan.

Browser → Access → hosted app → SQLite



Access gate

Cloudflare Access email OTP in front of the hostname.
Demo app login is separate (`operator / vc-demo`).

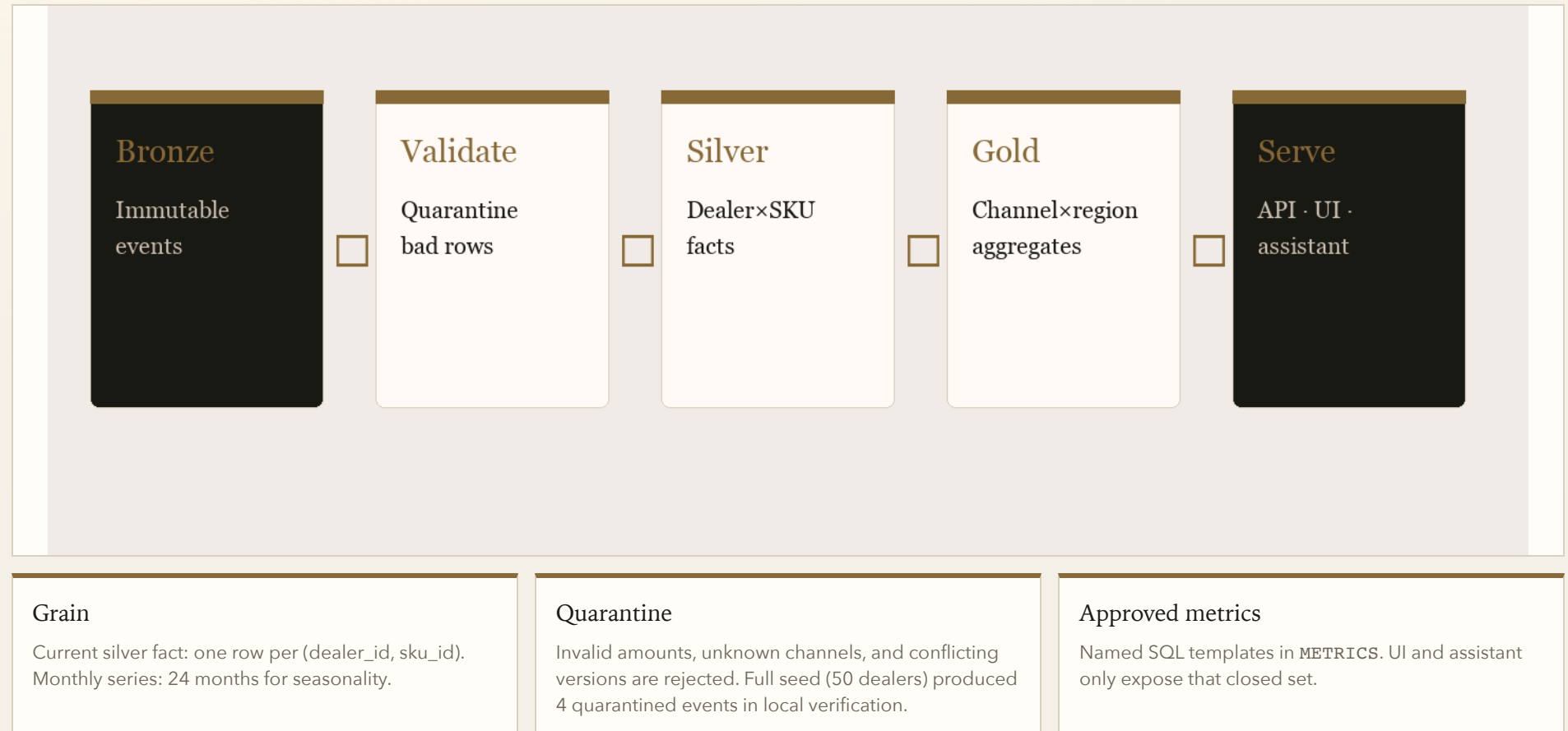
Hosted demo

Fly.io app `vc-retail-analytics` (ord) runs Python + tunnel. Observed machine state: started at verification time.

Session scope

HMAC-signed cookie binds dealer or operator role. Dealer sessions cannot request another dealer via query or body.

| Sources → silver facts → served metrics



Engineering checks and observed results

Local checks: c24bebd, 2026-10-02T20:38Z (re-run PASS). Hosted smoke: c24bebd image, 2026-10-02T20:45Z, PASS. Details in docs/VC_Submission_Verification.md.

CHECK	OBSERVED RESULT	EVIDENCE
Unit / API suite	PASS	6/6 tests in <code>tests/test_core.py</code> + <code>tests/test_assistant.py</code>
Dealer isolation	PASS	DLR-0001 session → GET <code>summary?dealer=DLR-0002</code> returns 403; ask with other dealer denied; portfolio rows only DLR-0001
Auth failure	PASS	Wrong password → 403
Metric reconcile	PASS	API summary net/margin matched independent SQL on <code>silver_facts</code> (tolerance \$0.02)
Unsupported forecast	PASS	Local: 400 clarify. Hosted: HTTP 400 (no fabricated rows)
Quarantine seed	PASS	Full seed quarantined=4 invalid fixture events
Snowflake parity	Not run	Adapter and SQL exist; no reconcile evidence in this pass
Hosted authenticated walkthrough	PASS	Operator path on <code>vc.datasharkbi.com</code> : Overview, Analytics (5 sections), assistant metric + SQL trace, catalog search, logout

| What is running vs proposed

COMPONENT	STATUS	NOTES
Hosted app + SQLite	Running	Fly machine started; Access OTP gate; authenticated hosted walkthrough PASS (see verification notes)
Deterministic assistant	Implemented + tested	Exact-match router; SQL trace; forecast refusal tested
Session dealer scope	Implemented + tested	Server-side scope + isolation tests
Analytics depth (Market + depth)	Implemented	Market forecast / competitors / GM / growth; margin / season / field / supply / procure / plan
Interview Lab	Implemented	Board brief, golden evals, Cortex vs governed, operator scope pin, 90-day plan
Voice on Snowflake	Implemented	Continuous Mic + board/portfolio brief; ElevenLabs; Snowflake-only numbers
Snowflake adapter / RAP	Verified	Hosted toggle + RAP evidence; Lab scope mirrors dealer pin
LLM as free-form answerer	Not the product	Cortex is discovery/fallback; production answers stay governed

Public support and authenticated dealer analytics serve different needs. This prototype explores how controlled warehouse metrics could support dealer and internal questions alongside familiar FAQ and escalation workflows. It is not an assessment of Visual Comfort internal systems.

Interview Lab — what to open with Rahul

1 • Board brief

Chains share expansion → GM lift → forecast vs run-rate → growth into one spoken narrative with SQL traces.

2 • Eval harness

20 golden prompts (FAQ · metric · voice fuzzy · refuse). Pass/fail + latency on screen.

3 • Cortex vs governed

Same question side-by-side. Prefer closed metrics; Cortex is discovery/fallback on Snowflake.

4 • Scope / RAP story

As `operator`, pin DLR-0001 – same ask, fewer rows. App-layer mirror of Snowflake RAP.

5 • What I'd ship in 90 days

Days 1-30 grounding · 31-60 assist + measure · 61-90 harden. Non-goals: inventing order status, replacing phone lanes day one.

Also: Market tab (competitors / GM / growth) and continuous Mic on Assistant. Login: `operator` / `vc-demo`.

| How to review the hosted demo

Access

- Open <https://vc.datasharkbi.com>
- Complete Cloudflare Access email OTP (allowlisted addresses only)
- Demo login: `operator / vc-demo`
- Optional tenancy check: `d1r-0001 / vc-demo`

Path to try

- Analytics → **Market**
- **Lab**: board brief → evals → Compare → pin DLR-0001
- Assistant Mic: "show share expansion"
- Refuse check: invent a forecast

Fallback if Access blocks you

Screenshots in this PDF are from the current synthetic build. Source is public at <https://github.com/prendleman/vc-retail-analytics>. Local run: `python3 -m app.server` then `http://127.0.0.1:8770/login`.

Hard-refresh after deploy to pull versioned assets.

CLOSE

What this prototype demonstrates

Synthetic dealer analytics with approved metrics, session scope, Snowflake toggle, voice, Market/competitor plays, and an Interview Lab for grounding, evals, and a 90-day plan – without inventing warehouse answers.

Limitations: synthetic data; competitor \$ are public-estimate calibrated; Access OTP required; capture rates are illustrative.

Open hosted demo →

<https://vc.datasharkbi.com> github.com/prendleman/vc-retail-analytics

Paul Rendleman

| Snowflake path (live on hosted toggle)

One UI targets SQLite or Snowflake. Hosted demo defaults to SQLite with a header switch. SERVING views, RAP, and Cortex Analyst are demonstrated; competitor metrics use dialect-safe CTE literals.

Present

Platform, transform, governance, semantic SQL; dual-backend flag; Lab Cortex compare.

Evidence

docs/evidence RAP dual-principal + Cortex Analyst; hosted health reports cortex + snowflake.

Honesty

Competitor revenues are public-estimate / assumption calibrated – not filed financials.

| Approved metric vocabulary (excerpt)

Market / growth / competitors

- demand_outlook, forecast_vs_runrate, growth_momentum, channel_growth
- competitor_landscape, competitive_position, share_expansion, region_expansion
- gm_opportunity_usd, discount_drag, channel_share

Core / margin / season / field

- portfolio, by_channel, margin_pct, margin_waterfall, seasonal_index
- territory_perf, whitespace, rep_grade, vendor_scorecard
- stock_risk, reorder_candidates, forecast_accuracy

Full definitions live in `app/core.py` (`METRICS` / `METRIC_PHRASES`). Assistant accepts only exact phrases from that map.